

NON-DESTRUCTIVE TESTING (NDT) INSPECTION REPORT

SovereignForge Industrial Compliance Asset Management

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DOCUMENT ID: INSP/2026/HX-204-Q3 | DATE: 15-JUN-2026 | CLASSIFICATION: CONFIDENTIAL

EQUIPMENT TARGET: CDU-03 COMPLEX - HEAT EXCHANGER HX-204 (SHELL & TUBE)

INSPECTION METHOD: ULTRASONIC NDT THICKNESS SURVEY (ASME SECTION V)

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1. EXECUTIVE MEASUREMENT SUMMARY:

- Equipment Nominal Design Thickness: 12.5 mm (ASME Section VIII Div 1)
- Statutory Minimum Retirement Thickness: 7.8 mm (SOP-4.2.1 Section 6.1)
- Minimum Measured Wall Thickness: 8.2 mm (Location: Nozzle N2 vicinity, Grid C4)
- Calculated Historical Corrosion Rate: 0.28 mm/year
- Calculated Remaining Service Life: 1.43 Years (approx. 17 months)

2. ULTRASONIC THICKNESS GRID MEASUREMENTS:

Grid A1: 10.4 mm | Grid A2: 10.1 mm | Grid A3: 9.8 mm

Grid B1: 9.5 mm | Grid B2: 9.2 mm | Grid B3: 8.9 mm

Grid C1: 8.8 mm | Grid C2: 8.5 mm | Grid C4 (Nozzle N2): 8.2 mm [CRITICAL THINNING]

3. FIELD NDT RECOMMENDATION (NON-CONFORMANCE DETECTED):

- Field inspector suggests deferring turnaround to 18 months (March 2028).
- WARNING: SOP-4.2.1 Section 7.1 strictly limits turnaround deferral to 12 months for sour crude service.
- Mandatory Action: Advance turnaround window to Q1 2027 to prevent containment breach.

REPORT SIGN-OFF: NDT Level-III Inspector Rajesh Kumar | Certified ISO-9712